

Contact

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(Portfolio)

Top Skills

Statistical Data Analysis

Data Science

Data Intelligence

Honors-Awards

Lester Touby Kurtz scholarship in graduate physics

Wai-Yim Ching and Mon Yin Lung scholarship in physics

Publications

Electronic structure, mechanical and optical properties of TiAl_3 (L1 2 & D0 22) via first-principles calculations

In search of zero thermal expansion anisotropy in Mo_5Si_3 by strategic alloying

Nature of interatomic bonding in controlling the mechanical properties of calcium silicate hydrates

Electronic structure and optical properties of amorphous GeO_2 in comparison to amorphous SiO_2

Building better water models using the shape of the charge distribution of a water molecule

Chamila Dharmawardhana

Data Scientist

United States

Summary

Data scientist and computational chemist with diverse experience in;

- Supervised and unsupervised learning models for real world applications
- Convolutional Neural Networks for image analysis
- Data cleaning and visualizations
- Ab-initio, molecular dynamics and coarse-grained simulation of technologically important materials
- proficiency in simulation and data analysis packages such as CHARMM, VASP, Gaussian, PotFit
- development of simulation methods data analysis techniques added to packages such as CHARMM, OLCAO
- excellent team player with aptitude for planning and conducting interdisciplinary projects

Skills & experience

languages: Python, TSQL, C++, Java, Fortran, BASH, Perl, Mathematica

frameworks: Anaconda, SciKit-learn, Pandas, numpy, scipy, TensorFlow, Keras, pytorch, BeautifulSoup, GitHub

HPCC: supercomputers (NERSC, XSEDE, ALCF). PBS, LSF and SLURM batch queuing systems

Management: Designed interdisciplinary research projects with diverse research collaborations effectively

Experience

Nordic Global

Data Scientist

January 2024 - Present (2 years 7 months)

United States

Bon Secours Mercy Health

Data Scientist

January 2022 - December 2023 (2 years)

United States

Promoboxx

Associate Data Scientist

November 2020 - January 2022 (1 year 3 months)

Boston, Massachusetts, United States

Illinois Institute of Technology

Senior Research Associate

2019 - 2020 (1 year)

Chicago, Illinois, United States

Development FFT based algorithms for protein docking

- Development of FFT based method (python, numpy, pandas) to evaluate binding affinity of protein complexes
- building scripts to streamline docking simulations of large databases
- Assist in grant proposal writing

Georgetown University

Postdoctoral Researcher

2017 - 2019 (2 years)

Washington, District of Columbia, United States

Development of efficient computational methods for biomolecular simulations

- Expand and improve SSMP water model
- Investigate current techniques for existing and ever expanding water models to improve and incorporate in SSMP water model
- added correction potential terms to CHARMM package to more accurately simulate ionic solutions
- Point person to collaborate with Laboratory of Computational Biology at NIH for coding potential function for the existing SSMP model and other projects
- Investigate diffusion of Ionic, Zwitterionic, and Polar solute in aqueous Solutions
- Trouble shooting and maintaining group workloads with HPC systems at XSEDE and NIH
- Substitute teacher: graduate level computational biology and undergraduate physical chemistry courses as needed
- Assisted in grant proposal and reviewing manuscripts

University of Colorado Boulder

Postdoctoral Fellow

2016 - 2017 (1 year)

Boulder, Colorado, United States

Force field building to study high temperature Mi-Si-B super-alloys

- Effectively collaborate with 8 PI collaboration in five universities under an ONR grant (MURI)
- Point person to participate monthly MURI Grad Student/Post Doc meetings to review and plan progress
- Graduate student mentoring
- Build force fields oxidation mechanisms of A15, C40 and T1/T2 phases at high temperature that are efficient enough for multimillion atom simulations.
- Build polarized force fields for BCC, FCC and HCP metals to mimic electronic structure Trouble shooting and maintaining group workloads with ALCF supercomputer local HPC cluster at CUB

University of Missouri-Kansas City

Graduate Research Assistant

2011 - 2015 (4 years)

Kansas City, Missouri, United States

DFT and AIMD methods to study complex ceramics, glass and metallic alloys

1. Utilized DFT methods to gain insight into structure at atomistic level of inorganic crystals

- Systematically survey large set of known cement minerals to identify important bonding in cements
- Discovered the effect of hydrogen bonding in cements and its impact on the mechanical properties
- Combinatorial study to identify stable structures for hydrogrossler mineral series Comparative study of germania and silica glass to understand their optical properties

2. High temperature thermomechanical properties of Mo-Si-B super-alloys and metallic glasses using ab-initio MD

- Developed methods to predict mechanical properties of Mo-Si-B alloys using a combination of AIMD and stress-strain method
- Used combinatorial approach to build Mo-Si-B with Al alloying in order to reduce the mechanical anisotropy of T1/T2 phase
- Discovered interesting elastic relations for all ternary MAX phases using a large database developed at our lab
- Investigate building classical potentials via force matching technique for selected MAX phases for various projects in our lab

3. Troubleshoot software, manage group accounts in supercomputers and mentor students
- Responsible for handling issues with software and user accounts in supercomputers (NERSC, XSEDE) and local HPCC clusters
 - Designed and function in interdisciplinary research projects with diverse research collaborations effectively
 - Assisted in grant proposal and quarterly progress report writing, reviewing manuscript Mentored three master's students, one undergraduate student, and one visiting scholar
 - Substitute teacher: graduate level mathematical physics and solid state courses as needed

Central Michigan University
Graduate Research And Teaching Assistant
2006 - 2011 (5 years)
Mount Pleasant, Michigan, United States

- Optical characterization of chalcogenide glasses via DFT gas phase calculations
- Calculated Raman, IR, NMR, NQR spectra of chalcogenide glasses clusters to identify unique structural units to understand the experimental spectra
 - Maintained Beowulf computer cluster in physics department
 - Teaching responsibilities
 - Instructed undergraduate physics laboratory
 - Tutored undergraduate students
 - Designed laboratory experiments and prepared student manual

University of Colombo
Teaching Assistant
2005 - 2006 (1 year)
Colombo, Western, Sri Lanka

Education

University of Missouri-Kansas City
Doctor of Philosophy - PhD, computational physics · (2011 - 2015)

Flatiron School
Data Science · (2020 - 2020)

Central Michigan University

Master of Science - MS, Theoretical and Mathematical Physics · (2006 - 2008)

University of Colombo

Bachelor of Science - BS, Physics and mathematics · (2001 - 2005)

National institute of business management, Sri Lanka

Diploma, Computer systems design · (2000 - 2001)